

Manufacturer	Customer	Business Transaction	
Advanced Steel Technologies GmbH Industriestraße 15 52070 Aachen DE quality@advanced-steel.de	Precision Engineering Ltd. Manufacturing Park 42 Birmingham B1 1AA GB procurement@precision-eng.co.uk	Order	PO-2024-7890
		Delivery	DN-2024-3456 · 500 kg

Product			
Product Name	Hardenability Test Steel	Shape	RoundBar - D 100 mm - Length 1000 mm
Batch ID	HTB-2024-045		

Chemical Analysis							
Heat Number H-2024-078 - Sample Location Ladle							
Symbol	C	Mn	Si	P	S	Cr	Ni
Unit	%	%	%	%	%	%	%
Min	0.42	0.50					
Max	0.50	0.80	0.40	0.030	0.025	0.25	0.25
Actual	0.45	0.65	0.25	0.015	0.008	0.12	0.08

Mechanical Properties										
Property		Symbol	Actual		Minimum		Maximum		Method	Status
Jominy Hardenability Test Quenched from 850°C									ISO 377	✓
Distance from quenched end (mm)	1.5	3	5	7	9	11	15	20	25	30
Min	42	39	35	32	29	26	22	20		
Max	47	46	44	41	39	37	33	31	<= 30	<= 29
Value [HRC]	45	43	40	36	33	31	27	25	28	27
Status	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Coating Thickness Profile				ISO 2178	✓
Magnetic induction method					
Position along length (mm)	0		50	100	
Min	>= 120		>= 110	>= 100	
Max	<= 130		<= 125	<= 120	
Value [µm]	125		115	108	
Status	Pass		Pass	Pass	

Validation	
We hereby certify that the material described above has been manufactured and tested in accordance with EN 10083-1 and ISO 377. All hardenability requirements have been met.	Validated by Dr. Maria Schmidt , Metallurgical Engineer, Quality Control Laboratory - 2024-12-15